VOLUME 22 | NUMBER 1 | WINTER 2010

Journal of APPLIED CORPORATE FINANCE

A MORGAN STANLEY PUBLICATION

In This Issue: Honoring Michael Jensen

Baylor University Roundtable on The Corporate Mission, CEO Pay, and Improving the Dialogue with Investors	8	*Panelists: Michael Jensen, Harvard Business School; Ron Naples, Quaker Chemical Corporation; Trevor Harris, Columbia University; and Don Chew, Morgan Stanley. Moderated by John Martin, Baylor University.*
Value Maximization, Stakeholder Theory, and the Corporate Objective Function	32	*Michael Jensen, Harvard Business School*
The Modern Industrial Revolution, Exit, and the Failure of Internal Control Systems	43	*Michael Jensen, Harvard Business School*
Just Say No to Wall Street: Putting a Stop to the Earnings Game	59	*Joseph Fuller, Monitor Group, and Michael Jensen, Harvard Business School*
CEO Incentives—It's Not How Much You Pay, But How	64	*Michael Jensen, Harvard Business School, and Kevin Murphy, University of Southern California*
Active Investors, LBOs, and the Privatization of Bankruptcy	77	*Michael Jensen, Harvard Business School*
Venture Capital in Canada: Lessons for Building (or Restoring) National Wealth	86	*Reuven Brenner, McGill University, and Gabrielle A. Brenner, HEC Montreal*
How to Tie Equity Compensation to Long-Term Results	99	*Lucian Bebchuk and Jesse Fried, Harvard Law School*
Executive Compensation: An Overview of Research on Corporate Practices and Proposed Reforms	107	*Michael Faulkender, Dalida Kadyrzhanova, N. Prabhala, and Lemma Senbet, University of Maryland*
Promotion Incentives and Corporate Performance: Is There a Bright Side to "Overpaying" the CEO?	119	*Jayant Kale, Georgia State University, Ebru Reis, Bentley University, and Anand Venkateswaran, Northeastern University*
Are Incentives the Bricks or the Building?	129	*Ron Schmidt, University of Rochester*

Morgan Stanley

Value Maximization, Stakeholder Theory, and the Corporate Objective Function

by Michael C. Jensen, The Monitor Group and Harvard Business School[1]

In most industrialized nations today, economists, management scholars, policy makers, corporate executives, and special interest groups are engaged in a high-stakes debate over corporate governance. In some scholarly and business circles, the discussion focuses mainly on questions of policies and procedures designed to improve oversight of corporate managers by boards of directors. But at the heart of the current global corporate governance debate is a remarkable division of opinion about the fundamental purpose of the corporation. Much of the discord can be traced to the complexity of the issues and to the strength of the conflicting interests that are likely to be affected by the outcome. But also fueling the controversy are political, social, evolutionary, and emotional forces that we don't usually think of as operating in the domain of business and economics. These forces serve to reinforce a model of corporate behavior that draws on concepts of "family" and "tribe." And as I argue in this paper, this model is an anachronism—a holdover from an earlier period of human development that nevertheless continues to cause much confusion among corporate managers about what it is that they and their organizations are supposed to do.

At the level of the individual organization, the most basic issue of governance is the following. Every organization has to ask and answer the question: What are we trying to accomplish? Or, to put the same question in more concrete terms: How do we keep score? When all is said and done, how do we measure better versus worse?

At the economy-wide or social level, the issue is this: If we could dictate the criterion or objective function to be maximized by firms (and thus the performance criterion by which corporate executives choose among alternative policy options), what would it be? Or, to put the issue even more simply: How do we want the firms in our economy to measure their own performance? How do we want them to determine what is better versus worse?

Most economists would answer simply that managers must have a criterion for evaluating performance and deciding between alternative courses of action, and that the criterion should be maximization of the long-term market value of the firm. (And "firm value," by the way, means not just the value of the equity, but the sum of the values of *all* financial claims on the firm—debt, warrants, and preferred stock, as well as equity.) This Value Maximization proposition has its roots in 200 years of research in economics and finance.

The main contender to value maximization as the corporate objective is called "stakeholder theory." Stakeholder theory says that managers should make decisions that take account of the interests of *all* the stakeholders in a firm. Stakeholders include all individuals or groups who can substantially affect, or be affected by, the welfare of the firm—a category that includes not only the financial claimholders, but also employees, customers, communities, and government officials.[2] In contrast to the grounding of value maximization in economics, stakeholder theory has its roots in sociology, organizational behavior, the politics of special interests, and, as I will discuss below, managerial self-interest. The theory is now popular and has received the formal endorsement of many professional organizations, special interest groups, and governmental bodies, including the current British government.[3]

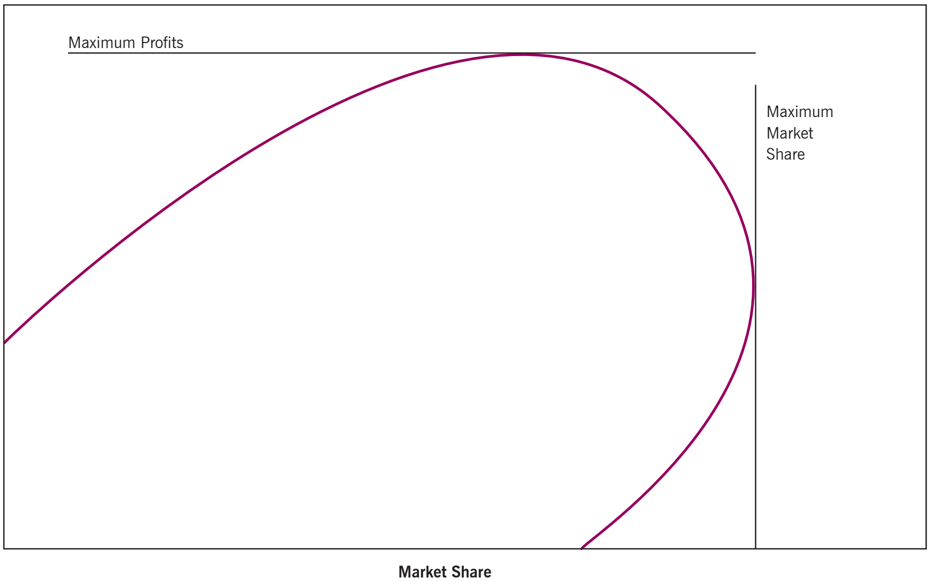
But, as I argue in this paper, stakeholder theory should not be viewed as a legitimate contender to value maximization because it fails to provide a *complete* specification of the corporate purpose or objective function. To put the matter more concretely, whereas value maximization provides corporate managers with a single objective, stakeholder theory directs corporate managers to serve "many masters." And, to

1. © 2001 Michael C. Jensen. An earlier version of this paper appears in *Breaking the Code of Change*, Michael Beer and Nithan Norhia, eds, Harvard Business School Press, 2000. This research has been supported by the The Monitor Group and Harvard Business School Division of Research. I am indebted to Nancy Nichols, Pat Meredith, Don Chew, and Janice Willett for many valuable suggestions.

2. Under some interpretations, stakeholders also include the environment, terrorists, blackmailers, and thieves. Edward Freeman, for example, writes: "The…definition of 'stakeholder' [is] any group or individual who can affect or is affected by the achievement of an organization's purpose.…For instance, some corporations must count 'terrorist groups' as stakeholders." (Edward R. Freeman, *Strategic Management: A Stakeholder Approach*, Pittman Books Limited, 1984, p. 53.)

3. See, for example, *Principles of Stakeholder Management: The Clarkson Principles*: The Clarkson Centre for Business Ethics, Joseph L. Rotman School of Management, Univ. of Toronto, Canada. For a critical analysis of stakeholder theory, I especially recommend the following articles by Elaine Sternberg: "Stakeholder Theory Exposed," *The Corporate Governance Quarterly* 2, no. 1 (1996); "The Stakeholder Concept: A Mistaken Doctrine," London: Foundation for Business Responsibilities, Issue Paper No.4 (November, 1999) (also available from the Social Science Research Network at: http://papers.ssrn.com/paper=263144). See also Sternberg's recent book, *Just Business: Business Ethics in Action*: Oxford University Press, 2000, which surveys the acceptance of stakeholder theory by the Business Roundtable and the *Financial Times*, and its recognition by law in 38 American states. On the latter issue, see also James L. Hanks, "From the Hustings: The Role of States with Takeover Control Laws." *Mergers & Acquisitions* 29, no. 2 (1994), September-October.

Figure 1 **Tradeoff Between Profits and Market Share**



paraphrase the old adage, when there are many masters, all end up being shortchanged. Without the clarity of mission provided by a single-valued objective function, companies embracing stakeholder theory will experience managerial confusion, conflict, inefficiency, and perhaps even competitive failure. And the same fate is likely to be visited on those companies that use the so-called "Balanced Scorecard" approach—the managerial equivalent of stakeholder theory—as a performance measurement system.

But if stakeholder theory and the Balanced Scorecard can destroy value by obscuring the overriding corporate goal, does that mean they have no legitimate corporate uses? And can corporate managers succeed by simply holding up value maximization as the goal and ignoring their stakeholders? The answer to both is an emphatic *no*. In order to maximize value, corporate managers must not only satisfy, but enlist the support of, all corporate stakeholders—customers, employees, managers, suppliers, local communities. Top management plays a critical role in this function through its leadership and effectiveness in creating, projecting, and sustaining the company's strategic vision. And even if the Balanced Scorecard is likely to be counterproductive as a performance evaluation and reward system, the *process* of creating the scorecard can add significant value by helping managers understand both the company's strategy and the drivers of value in their businesses.

With this in mind, I clarify what I believe is the proper relation between value maximization and stakeholder theory by proposing a (somewhat) new corporate objective function. I call it *enlightened* value maximization, and it is identical to what I call *enlightened* stakeholder theory. Enlightened value maximization uses much of the structure of stakeholder theory but accepts maximization of the long-run value of the firm as the criterion for making the requisite tradeoffs among its stakeholders. Enlightened stakeholder theory, while focusing attention on meeting the demands of all important corporate constituencies, specifies long-term value maximization as the firm's objective. In so doing, it solves the problems arising from the multiple objectives that accompany traditional stakeholder theory by giving managers a clear way to think about and make the tradeoffs among corporate stakeholders.

The Logical Structure of the Problem

In discussing whether firms should maximize value or not, we must separate two distinct issues:

1. Should the firm have a single-valued objective?

2. And, if so, should that objective be value maximization or something else (for example, maintaining employment or improving the environment)?

The debate over whether corporations should maximize value or act in the interests of their stakeholders is generally couched in terms of the second issue, and is often mistakenly framed as stockholders *versus* stakeholders. The real conflict here, though this is rarely stated or even recognized, is over the first issue—that is, whether the firm should have a single-valued objective function or scorecard. The failure to frame the problem in this way has contributed greatly to widespread misunderstanding and contentiousness.

What is commonly known as stakeholder theory, while not totally without content, is fundamentally flawed because it violates the proposition that a single-valued objective is a prerequisite for purposeful or rational behavior by any organization. In particular a firm that adopts stakeholder theory will be handicapped in the competition for survival because, as a basis for action, stakeholder theory politicizes the corporation and leaves its managers empowered to exercise their own preferences in spending the firm's resources.

Issue #1: Purposeful Behavior Requires the Existence of a Single-Valued Objective Function.

Consider a firm that wishes to increase both its current-year profits and its market share. Assume, as shown in Figure 1, that over some range of values of market share, profits increase. But, at some point, increases in market share come only at the expense of reduced current-year profits—say, because increased expenditures on R&D and advertising, or price reductions to increase market share, reduce this year's profit. Therefore, it is not logically possible to speak of maximizing both market share and profits.

In this situation, it is impossible for a manager to decide on the level of R&D, advertising, or price reductions because he or she is faced with the need to make tradeoffs between the two "goods"—profits and market share—but has no way to do so. While the manager knows that the firm should be at the point of maximum profits or maximum market share (or somewhere between them), there is no purposeful way to decide where to be in the area in which the firm can obtain more of one good only by giving up some of the other.

Multiple Objectives is No Objective

It is logically impossible to maximize in more than one dimension at the same time unless the dimensions are what are known as "monotonic transformations" of one another. Thus, telling a manager to maximize current profits, market share, future growth in profits, and anything else one pleases will leave that manager with no way to make a reasoned decision. In effect, it leaves the manager with *no* objective. The result will be confusion and a lack of purpose that will handicap the firm in its competition for survival.[4]

A company can resolve this ambiguity by specifying the tradeoffs among the various dimensions, and doing so amounts to specifying an overall objective such as $V = f(x, y, \ldots)$ that explicitly incorporates the effects of decisions on *all* the performance criteria—all the goods or bads (denoted by $(x, y, \ldots)$) that can affect the firm (such as cash flow, risk, and so on). At this point, the logic above does not specify what V is. It could be anything the board of directors chooses, such as employment, sales, or growth in output. But, as I argue below, social welfare and survival will severely constrain the boards' choices.

Nothing in the analysis so far has said that the objective function f must be well behaved and easy to maximize. If the function is non-monotonic, or even chaotic, it makes it more difficult for managers to find the overall maximum. (For example, as I discuss later, the relationship between the value of the firm and a company's current earnings and investors' expectations about its future earnings and investment expenditures will often be difficult to formulate with much precision.) But even in these situations, the meaning of "better" or "worse" is defined, and managers and their monitors have a "principled"—that is, an objective and theoretically consistent—basis for choosing and auditing decisions. Their choices are not just a matter of their own personal preferences among various goods and bads.

Given managers' uncertainty about the exact specification of the objective function, it is perhaps better to call the objective function "value seeking" rather than value maximization. This way one avoids the confusion that arises when some argue that maximizing is difficult or impossible if the world is structured in sufficiently complicated ways.[5] It is not necessary that we be able to maximize, only that we can tell when we are getting better—that is moving in the right direction.

Issue #2: Total Firm Value Maximization Makes Society Better Off

Given that a firm must have a single objective that tells us what is better and what is worse, we then must face the issue of what that definition of better is. Even though the single objective will always be a complicated function of many different goods or bads, the short answer to the question is that 200 years' worth of work in economics and finance indicate that social welfare is maximized when all firms in an economy attempt to maximize their own total firm value. The intuition behind this criterion is simple: that value is created—and when I say "value" I mean "social" value—whenever a firm produces an output, or set of outputs, that is valued by its customers at more than the value of the inputs it consumes (as valued by their suppliers) in the production of the outputs. Firm value is simply the long-term market value of this expected stream of benefits.

To be sure, there are circumstances when the value-maximizing criterion does not maximize social welfare—notably, when there are monopolies or "externali-

4. For a case study of a small non-profit firm that almost destroyed itself while trying to maximize over a dozen dimensions at the same time, see Michael Jensen, Karen H. Wruck, and Brian Barry, "Fighton, Inc. (A) and (B)," Harvard Business School Case #9-391-056, March 20, 1991; and Karen Wruck, Michael Jensen, and Brian Barry, "Fighton, Inc., (A) and (B) Teaching Note," Case #5-491-111, Harvard Business School, 1991. For an interesting empirical paper that formally tests the proposition that multiple objectives handicap firms, see Kees Cools and Mirjam van Praag (2000), "The Value Relevance of a Single-Valued Corporate Target: An Empirical Analysis." Available from the Social Science Research Network eLibrary at http://papers.ssrn.com/paper=244788. In their test using 80 Dutch firms in the 1993-1997 period, the authors conclude: "Our findings show the importance of setting one single target for value creation." (emphasis in original)

5. I'd like to thank David Rose for suggesting this simple and more descriptive term for value maximizing. See David C. Rose, "Teams, Firms, and the Evolution of Profit Seeking Behavior," May, 1999, Dept. of Economics, University of Missouri-St. Louis, St. Louis, MO, Unpublished Manuscript, available from the Social Science Research Network eLibrary at: http://papers.ssrn.com/paper=224438.

ties." Monopolies tend to charge prices that are too high, resulting in less than the socially optimal levels of production. By "externalities," economists mean situations in which decision-makers do not bear the full cost or benefit consequences of their choices or actions. Examples are cases of air or water pollution in which a firm adds pollution to the environment without having to purchase the right to do so from the parties giving up the clean air or water.

But there can be no externalities as long as alienable property rights in all physical assets are defined and assigned to some private individual or firm. Thus, the solution to these problems lies not in telling firms to maximize something other than profits, but in defining and then assigning to some private entity the alienable decision rights necessary to eliminate the externalities.[6] In any case, resolving externality and monopoly problems, as I will discuss later, is the legitimate domain of the government in its rule-setting function.[7]

Maximizing the total market value of the firm—that is, the sum of the market values of the equity, debt and any other contingent claims outstanding on the firm—is the objective function that will guide managers in making the optimal tradeoffs among multiple constituencies (or stakeholders). It tells the firm to spend an additional dollar of resources to satisfy the desires of each constituency as long as that constituency values the result at more than a dollar. In this case, the payoff to the firm from that investment of resources is at least a dollar (in terms of market value). Although there are many single-valued objective functions that could guide a firm's managers in their decisions, value maximization is an important one because it leads under most conditions to the maximization of social welfare. But let's look more closely at this.

Value Maximizing and Social Welfare

Much of the discussion in policy circles about the proper corporate objective casts the issue in terms of the conflict among various constituencies, or "stakeholders," in the corporation. The question then becomes whether shareholders should be held in higher regard than other constituencies, such as employees, customers, creditors, and so on. But it is both unproductive and incorrect to frame the issue in this manner. The real issue is what corporate behavior will get the most out of society's limited resources—or equivalently, what behavior will result in the least social waste—not whether one group is or should be more privileged than another.

Profit Maximization: A Simplified Case

To see how value maximization leads to a socially efficient solution, let's first consider an objective function, profit maximization, in a world in which all production runs are infinite and cash flow streams are level and perpetual. This scenario with level and perpetual streams allows us to ignore the complexity introduced by the tradeoffs between current and future-year profits (or, more accurately, cash flows).

Consider now the social welfare effects of a firm's decision to take resources out of the economy in the form of labor hours, capital, or materials purchased voluntarily from their owners in single-price markets. The firm uses these inputs to produce outputs of goods or services that are then sold to consumers through voluntary transactions in single-price markets.

In this simple situation, a company that takes inputs out of the economy and puts its output of goods and services back into the economy increases aggregate welfare if the prices at which it sells the goods more than cover the costs it incurs in purchasing the inputs (including, of course, the cost of the capital the firm is using). Clearly the firm should expand its output as long as an additional dollar of resources taken out of the economy is valued by the consumers of the incremental product at more than a dollar. Note that it is precisely because profit is the amount by which revenues exceed costs—by which the value of output exceeds the value of inputs—that profit maximization[8] leads to an efficient social outcome.[9]

Because the transactions are voluntary, we know that the owners of the inputs value them at a level less than or equal to the price the firm pays—otherwise they wouldn't sell them. Therefore, as long as there are no negative externalities in the input factor markets,[10] the opportunity cost to society of those inputs is no higher than the total cost to the firm of acquiring them. I say "no higher" because some suppliers of inputs to the firm are able to earn "rents" by obtaining prices higher than the value of the goods to them. But such rents do not represent social costs, only transfers of wealth to those suppliers. Likewise, as long as there are no externalities in the output markets, the value to society of the goods and services produced by the firm is at least as great as the price the firm receives for the sale of those goods and services. If this were not true, the individuals purchasing them would not do so. Again, as in the case of producer surplus on inputs, the benefit to society is higher to the extent that consumer surplus

6. See Ronald H. Coase, "The Problem of Social Cost," *Journal of Law and Economics* 3, no. October 1960: pp. 1-44; and Michael C. Jensen and William H. Meckling, "Specific and General Knowledge, and Organization Structure," in ed. Lars Werin and Hans Wijkander, *Contract Economics*: (Oxford: Basil Blackwell, 1992), pp. 251-274. Available from the Social Science Research Network eLibrary at: http://papers.ssrn.com/paper=6658.

7. In the case of a monopoly, profit maximization leads to a loss of social product because the firm expands production only to the point where an additional dollar's worth of inputs generates incremental revenues equal to a dollar, not where consumers value the incremental product at a dollar. In this case the firm produces less of a commodity than that which would result in maximum social welfare.

In addition, we should recognize that when a complete set of claims for all goods for each possible time and state of the world do not exist, the social maximum will be constrained; but this is just another recognition of the fact that we must take into account the costs of creating additional claims and markets in time/state delineated claims. See Kenneth J. Arrow, "The Role of Securities in the Optimal Allocation of Risk Bearing." *Review of Economic Studies* 31, no. 86 (1964): pp. 91-96; and Gerard Debreu, *Theory of Value* (New York: John Wiley & Sons, 1959).

8. Again, provided there are no externalities.

9. I am indebted to my colleague George Baker for this simple way of expressing the social optimality of profit maximization.

10. An example would be a case where the supplier of an input was imposing negative externalities on others by polluting water or air.

exists (that is, to the extent that some consumers are able to purchase the output at prices lower than the value to them).

In sum, when a company acquires an additional unit of any input(s) to produce an additional unit of any output, it increases social welfare by at least the amount of its profit—the difference between the value of the output and the cost of the input(s) required in producing it.[11] And thus the signals to the management are clear: Continue to expand purchases of inputs and sell the resulting outputs as long as an additional dollar of inputs generates sales of at least a dollar.

Value and Tradeoffs through Time

In a world in which cash flows, profits, and costs are not uniform over time, managers must deal with the tradeoffs of these items through time. A common case is when a company's capital investment comes in lumps that have to be funded up front, while production and revenue occurs in the future. Knowing whether society will be benefited or harmed requires knowing whether the future output will be valuable enough to offset the cost of having people give up their labor, capital, and material inputs in the present. Interest rates help us make this decision by telling us the cost of giving up a unit of a good today for receipt at some time in the future. So long as people take advantage of the opportunity to borrow or lend at a given interest rate, that rate determines the value of moving a marginal dollar of resources (inputs or consumption goods) forward or backward in time.[12] In this world, individuals are as well off as possible if they maximize their wealth as measured by the discounted present value of all future claims.

In addition to interest rates, managers also need to take into account the risk of their investments and the premium the market charges for bearing such risk. But, when we add uncertainty and risk into the equation, nothing of major importance is changed in this proposition as long as there are capital markets in which the individual can buy and sell risk at a given price. In this case, it is the risk-adjusted interest rate that is used in calculating the market value of risky claims. The corporate objective function that maximizes social welfare thus becomes "maximize current total firm market value." It tells firms to expand output and investment to the point where the present market value of the firm is at a maximum.[13]

Stakeholder Theory

To the extent that stakeholder theory says that firms should pay attention to all their constituencies, the theory is unassailable. Taken this far stakeholder theory is completely consistent with value maximization or value-seeking behavior, which implies that managers must pay attention to all constituencies that can affect the value of the firm.

But there is more to the stakeholder story than this. Any theory of corporate decision making must tell the decision-makers—in this case, managers and boards of directors—how to choose among multiple constituencies with competing and, in some cases, conflicting interests. Customers want low prices, high quality, and full service. Employees want high wages, high-quality working conditions, and fringe benefits, including vacations, medical benefits, and pensions. Suppliers of capital want low risk and high returns. Communities want high charitable contributions, social expenditures by companies to benefit the community at large, increased local investment, and stable employment. And so it goes with every conceivable constituency. Obviously any decision criterion—and the objective function is at the core of any decision criterion—must specify how to make the tradeoffs between these demands.

The Specification of Tradeoffs and the Incompleteness of Stakeholder Theory

Value maximization (or value seeking) provides the following answer to the tradeoff question: Spend an additional dollar on any constituency provided the long-term value added to the firm from such expenditure is a dollar or more. Stakeholder theory, by contrast,[14] contains no conceptual specification of how to make the tradeoffs among stakeholders. And as I argue below, it is this failure to provide a criterion for making such tradeoffs, or even to acknowledge the need for them, that makes stakeholder theory a prescription for destroying firm value and reducing social welfare. This failure also helps explains the theory's remarkable popularity.

Implications for Managers and Directors

Because stakeholder theory leaves boards of directors and executives in firms with no principled criterion for decision making, companies that try to follow the dictates of stakeholder theory will eventually fail if they are competing with firms that are aiming to maximize value. If this is true, why do so many managers and directors of corporations embrace stakeholder theory?

One answer lies in their personal short-run interests. By failing to provide a definition of better, stakeholder theory effectively leaves managers and directors unaccountable for their stewardship of the firm's resources. Without criteria for performance, managers cannot be evaluated in any principled way. Therefore, stakeholder theory plays into the hands of managers by allowing them to pursue their own interests at the expense of the firm's financial claimants and society at large. It allows managers and directors to devote the firm's

11. Equality holds only in the special case where consumer and producer surpluses are zero, and there are no externalities or monopoly.

12. For those unfamiliar with finance and present values, the value one year from now of a dollar today saved for use one year from now is thus \$1 x (1+r), where r is the interest rate. Alternatively, the value today of a dollar of resources to be received one year from now is its present value of \$1/(1+r).

13. Without going into the details here, the same criterion applies to all organizations whether they are public corporations or not. Obviously, even if the financial claims are not explicitly valued by the market, social welfare will be increased as long as managers of partnerships or non-profits increase output so long as the imputed market value of claims on the firm continue to increase.

14. At least as advocated by Freeman (1984), Clarkson Principles (1999) and others.

resources to their own favorite causes—the environment, art, cities, medical research—without being held accountable for the effect of such expenditures on firm value. (And this can be true even though managers may not consciously recognize that adopting stakeholder theory leaves them unaccountable—especially, for example when such managers have a strong personal interest in social issues.) By expanding the power of managers in this unproductive way, stakeholder theory increases agency costs in the economic system. And since it expands the power of managers, it is not surprising that stakeholder theory receives substantial support from them.

In this sense, then, stakeholder theory can be seen as gutting the foundations of the firm's internal control systems. By "internal control systems," I mean mainly the corporate performance measurement and evaluation systems that, when properly designed, provide strong incentives for value-increasing behavior. There is simply no principled way within the stakeholder construct (which fails to specify what better is) that anyone could say that a manager has done a good or bad job. Stakeholder theory supplants or weakens the power of such control systems by giving managers more power to do whatever they want, subject only to constraints that are imposed by forces *outside* the firm—by the financial markets, the market for corporate control (e.g., the market for hostile takeovers), and, when all else fails, the product markets.

Thus, having observed the efforts of stakeholder theory advocates to weaken internal control systems, it is not surprising to see the theory being used to argue for government restrictions, such as state anti-takeover provisions, on financial markets and the market for corporate control. These markets are driven by value maximization and will limit the damage that can be done by managers who adopt stakeholder theory. And, as illustrated by the 1990s campaigns against globalization and free trade, the stakeholder argument is also being used to restrict product-market competition as well.

But there is something deeper than self-interest—something rooted in the evolution of the human psyche—that is driving our attraction to stakeholder theory.

Families Versus Markets: The Roots of Stakeholder Theory

Stakeholder theory taps into the deep emotional commitment of most individuals to the family and tribe. For tens of thousands of years, those of our ancestors who had little respect for or loyalty to the family, band, or tribe were much less likely to survive than those who did. In the last few hundred years, we have experienced the emergence of a market exchange system of prices and the private property rights on which they are based. This system of voluntary and decentralized coordination of human action has brought huge increases in human welfare and freedom of action.

As Friedrich von Hayek points out, we are generally unaware of the functioning of these market systems because no single mind invented or designed them—and because they work in very complicated and subtle ways. In Hayek's words:

> *We are led—for example, by the pricing system in market exchange—to do things by circumstances of which we are largely unaware and which produce results that we do not intend. In our economic activities we do not know the needs which we satisfy nor the sources of the things which we get. Almost all of us serve people whom we do not know, and even of whose existence we are ignorant; and we in turn constantly live on the services of other people of whom we know nothing. All this is possible because we stand in a great framework of institutions and traditions—economic, legal, moral—into which we fit ourselves by obeying certain rules of conduct that we never made, and which we have never understood in the sense in which we understand how the things that we manufacture function.*[15]

Moreover, these systems operate in ways that limit the options of the small group or family, and these constraints are neither well understood nor instinctively welcomed by individuals. Many people are drawn to stakeholder theory through their evolutionary attachment to the small group and the family. As Hayek puts it:

> *Constraints on the practices of the small group, it must be emphasized and repeated, are hated. For, as we shall see, the individual following them, even though he depends on them for life, does not and usually cannot understand how they function or how they benefit him. He knows so many objects that seem desirable but for which he is not permitted to grasp, and he cannot see how other beneficial features of his environment depend on the discipline to which he is forced to submit—a discipline forbidding him to reach out for these same appealing objects. Disliking these constraints so much, we hardly can be said to have selected them; rather, these constraints selected us: they enabled us to survive.*[16]

Thus we have a system in which human beings must simultaneously exist in two orders, what Hayek calls the "micro-cosmos" and the "macro-cosmos":

> *Moreover, the structures of the extended order are made up not only of individuals but also of many, often overlapping, suborders within which old instinctual responses, such as solidarity and altruism, continue to retain some importance by assisting voluntary collaboration, even though they are incapable, by themselves, of creating a basis for the more extended order. Part of our present difficulty is that we must constantly adjust*

15. F. A. Hayek, *The Fatal Conceit*. Edited by W. W. Bartley. The Collected Works of F. A. Hayek. Chicago: University of Chicago Press, 1988, p.14.

16. Ibid., pp. 13, 14; emphasis in original.

our lives, our thoughts and our emotions, in order to live simultaneously within different kinds of orders according to different rules. If we were to apply the unmodified, uncurbed rules of the micro-cosmos (i.e. of the small band or troop, or of, say, our families) to the macro-cosmos (our wider civilization), as our instincts and sentimental yearnings often make us wish to do, we would destroy it. Yet if we were always to apply the rules of the extended order to our more intimate groupings, we would crush them. So we must learn to live in two sorts of worlds at once. To apply the name "society" to both, or even to either, is hardly of any use, and can be most misleading.[17]

Stakeholder theory taps into this confusion and antagonism towards markets and relaxes constraints on the small group in ways that are damaging to society as a whole and (in the long run) to the small group itself. Such deeply rooted and generally unrecognized conflict between allegiances to family and tribe and what is good for society as a whole has had a major impact on our evolution. And in this case, the conflict does not end up serving our long-run collective interests.[18]

Enlightened Value Maximization and Enlightened Stakeholder Theory

For those intent on improving management, organizational governance, and performance, there is a way out of the conflict between value maximizing and stakeholder theory. It lies in the melding together of what I call "enlightened value maximization" and "enlightened stakeholder theory."

Enlightened Value Maximization

Enlightened value maximization recognizes that communication with and motivation of an organization's managers, employees, and partners is extremely difficult. What this means in practice is that if we simply tell all participants in an organization that its sole purpose is to maximize value, we will not get maximum value for the organization. Value maximization is not a vision or a strategy or even a purpose; it is the scorecard for the organization. We must give people enough structure to understand what maximizing value means so that they can be guided by it and therefore have a chance to actually achieve it. They must be turned on by the vision or the strategy in the sense that it taps into some human desire or passion of their own—for example, a desire to build the world's best automobile or to create a film or play that will move people for centuries. All this can be not only consistent with value seeking, but a major contributor to it.

And this brings us up against the limits of value maximization *per* se. Value seeking tells an organization and its participants how their success in achieving a vision or in implementing a strategy will be assessed. But value maximizing or value seeking says nothing about how to create a superior vision or strategy. Nor does it tell employees or managers how to find or establish initiatives or ventures that create value. It only tells them how we will measure success in their activity.

Defining what it means to score a goal in football or soccer, for example, tells the players nothing about how to win the game; it just tells them how the score will be kept. That is the role of value maximization in organizational life. It doesn't tell us how to have a great defense or offense, or what kind of plays to create, or how much to train and practice, or whom to hire, and so on. All of these critical functions are part of the competitive and organizational strategy of any team or organization. Adopting value creation as the scorekeeping measure does nothing to relieve us of the responsibility to do all these things and more in order to survive and dominate our sector of the competitive landscape.

This means, for example, that we must give employees and managers a structure that will help them resist the temptation to maximize short-term financial performance (as typically measured by accounting profits or, even worse, earnings per share). Short-term profit maximization at the expense of long-term value creation is a sure way to destroy value. This is where enlightened stakeholder theory can play an important role. We can learn from stakeholder theorists how to lead managers and participants in an organization to think more generally and creatively about how the organization's policies treat all important constituencies of the firm. This includes not just the stockholders and financial markets, but employees, customers, suppliers, and the community in which the organization exists.

Indeed, it is a basic principle of enlightened value maximization that *we cannot maximize the long-term market value of an organization if we ignore or mistreat any important constituency*. We cannot create value without good relations with customers, employees, financial backers, suppliers, regulators, and communities. But having said that, we can now use the value criterion for choosing among those competing interests. I say "competing" interests because no constituency can be given full satisfaction if the firm is to flourish and survive. Moreover, we can be sure—again, apart from the possibility of externalities and monopoly power—that using this value criterion will result in making society as well off as it can be.

17. Ibid., p. 18; emphasis in original.

18. It is useful here to briefly summarize the positive arguments (those refutable by empirical data) and normative arguments (those propositions that say what should be rather than what is in the world) I have made thus far. I have argued positively that firms that follow stakeholder theory as it is generally advocated will do less well in the competition for survival than those who follow a well-defined single-valued objective such as value creation. I have also argued positively that if firms follow value creation, social welfare will be greater and normatively that this is desirable. I have also argued positively that the self-interests of managers and directors will lead them to prefer stakeholder theory because it increases their power and means they cannot be held accountable for their actions. I have also argued positively that the self-interest of special interest groups who wish to acquire legitimacy in corporate governance circles to enhance their influence over the allocation of corporate resources will advocate the use of stakeholder theory by managers and directors. This leads to the positive prediction that society will be poorer if they are successful, and to the normative conclusion that this is undesirable. For a discussion of the role of normative, positive (or instrumental), and descriptive theory in the literature on stakeholder theory, see Thomas Donaldson and Lee E. Preston, "The Stakeholder Theory of the Corporation: Concepts, Evidence, and Implications." *Academy of Management Review 20*, no. 1 (1995): pp. 65-91.

As stated earlier, resolving externality and monopoly problems is the legitimate domain of the government in its rule-setting function. Those who care about resolving monopoly and externality issues will not succeed if they look to corporations to resolve these issues voluntarily. Companies that try to do so either will be eliminated by competitors who choose not to be so civic minded, or will survive only by consuming their economic rents in this manner.

Enlightened Stakeholder Theory

Enlightened stakeholder theory is easy to explain. It can make use of most of what stakeholder theorists offer in the way of processes and audits to measure and evaluate the firm's management of its relations with all important constituencies. Enlightened stakeholder theory adds the simple specification that the objective function—the overriding goal—of the firm is to maximize total long-term firm market value. In short, the change in the total long-term market value of the firm is the scorecard by which success is measured.

I say "long-term" market value to recognize the possibility that financial markets, although forward-looking, may not understand the full implications of a company's policies until they begin to show up in cash flows over time. In such cases, management must communicate to investors the policies' anticipated effect on value, and then wait for the market to catch up and recognize the real value of its decisions as reflected in increases in market share, customer and employee loyalty, and, finally, cash flows. Value creation does not mean responding to the day-to-day fluctuations in a firm's value. The market is inevitably ignorant of many managerial actions and opportunities, at least in the short run. In those situations where the financial markets clearly do not have this private competitive information, directors and managers must resist the pressures of those markets while making every effort to communicate their expectations to investors.

In this way, enlightened stakeholder theorists can see that although stockholders are not some special constituency that ranks above all others, long-term stock value is an important determinant (along with the value of debt and other instruments) of total long-term firm value. They would recognize that value creation gives management a way to assess the tradeoffs that must be made among competing constituencies, and that it allows for principled decision-making independent of the personal preferences of managers and directors. Also important, managers and directors become accountable for the assets under their control because the value scorecard provides an objective yardstick against which their performance can be evaluated.

Measurability and Imperfect Knowledge

It is important to recognize that none of the above arguments depends on value being easily observable. Nor do they depend on perfect knowledge of the effects on value of decisions regarding any of a firm's constituencies. The world may be complex and difficult to understand. It may leave us in deep uncertainty about the effects of any decisions we may make. It may be governed by complex dynamic systems that are difficult to optimize in the usual sense. But that does not remove the necessity of making choices on a day-to-day basis. And to do this in a purposeful way we must have a scorecard.

The absence of a scorecard makes it easier for people to engage in value-claiming activities that satisfy one or more group of stakeholders at the expense of value creation. We can take random actions, and we can devise decision rules that depend on superstitions. But none of these is likely to serve us well in the competition for survival.

We must not confuse optimization with value creation or value seeking. To create value we need not know exactly what maximum value is and precisely how it can be achieved. What we must do, however, is to set up our organizations so that managers and employees are clearly motivated to seek value—to institute those changes and strategies that are most likely to cause value to rise. To navigate in such a world in anything close to a purposeful way, we must have a notion of "better," and value seeking is such a notion. I know of no other scorecard that will score the game as well as this one. Under most circumstances and conditions, it tells us when we are getting better, and when we are getting worse. It is not perfect, but that is the nature of the world.

The Balanced Scorecard

The Balanced Scorecard is the managerial equivalent of stakeholder theory. Like stakeholder theory, the notion of a "balanced" scorecard appeals to many, but it suffers from many of the same flaws. When we use multiple measures on the balanced scorecard to evaluate the performance of people or business units, we put managers in the same impossible position as managers trying to manage under stakeholder theory. We are asking them to maximize in more than one dimension at a time with no idea of the tradeoffs between the measures. As a result, purposeful decisions cannot be made.

The balanced scorecard arose from the belief of its originators, Robert Kaplan and David Norton, that purely financial measures of performance are not sufficient to yield effective management decisions.[19]

I agree with this conclusion though, as I suggest below, they have inadvertently confused this with the unstated, but implicit conclusion that there should never be a *single* measure of performance. Moreover, especially *at lower levels of an organization*, a single pure financial measure of performance is unlikely to properly measure a person's or even a business unit's contribution to a company. In the words of Kaplan and Norton:

> *The Balanced Scorecard complements financial measures of past performance with measures of the drivers of future perfor-*

mance. The objectives and measures of the scorecard are derived from an organization's vision and strategy. The objectives and measures view organizational performance from four perspectives: financial, customer, internal business process, and learning and growth…The Balanced Scorecard expands the set of business unit objectives beyond summary financial measures. Corporate executives can now measure how their business units create value for current and future customers and how they must enhance internal capabilities and the investment in people, systems, and procedures necessary to improve future performance. The Balanced Scorecard captures the critical value-creation activities created by skilled, motivated organizational participants. While retaining, via the financial perspective, an interest in short-term performance, the Balanced Scorecard clearly reveals the value drivers for superior long-term financial and competitive performance.[20]

As Kaplan and Norton go on to say,

The measures are balanced *between the outcome measures—the results of past efforts—and the measures that drive future performance. And the scorecard is balanced between objective easily quantified outcome measures and subjective, somewhat judgmental performance drivers of the outcome measures…*

A good balanced scorecard should have an appropriate mix of outcomes (lagging indicators) and performance drivers (leading indicators) that have been customized to the business unit's strategy.[21]

The aim of Kaplan and Norton, then, is to capture both past performance and expected future performance in scorecards with multiple measures—in fact, as many as two dozen of them—that are intimately related to the organization's strategy.[22] And this is where my misgivings about the Balanced Scorecard lie. For an organization's strategy to be implemented effectively, each person in the organization must clearly understand what he or she has to do, how their performance measures will be constructed, and how their rewards and punishments are related to those measures.

But, as we saw earlier in the case of multiple constituencies (or the multiple goals represented in Figure 1), decision makers cannot make rational choices without some overall single dimensional objective to be maximized. Given a dozen or two dozen measures and no sense of the tradeoffs between them, the typical manager will be unable to behave purposefully, and the result will be confusion.

Kaplan and Norton generally do not deal with the critical issue of how to weight the multiple dimensions represented by the two-dozen measures on their scorecards. And this is where problems with the Balanced Scorecard are sure to arise: without specifying what the tradeoffs are among these two dozen or so different measures, there is no "balance" in their scorecard. Adding to the potential for confusion, Kaplan and Norton also offer almost no guidance on the critical issue of how to tie the performance measurement system to managerial incentives and rewards. Here is their concluding statement on this important matter:

Several approaches may be attractive to pursue. In the short term, tying incentive compensation of all senior managers to a balanced set of business unit scorecard measures will foster commitment to overall organizational goals, rather than suboptimization within functional departments…Whether such linkages should be explicit... or applied judgmentally… will likely vary from company to company. More knowledge about the benefits and costs of explicit linkages will undoubtedly continue to be accumulated in the years ahead.[23]

What the Balanced Scorecard fails to provide, then, is a clear linkage (and a rationale for that linkage) between the performance measures and the corporate system of rewards and punishments. Indeed, the Balanced Scorecard does not provide a scorecard in the traditional sense of the word. And, to make my point, let me push the sports analogy a little further. A scorecard in any sport yields a single number that determines the winner among all contestants. In most sports the person or team with the highest score wins. Very simply, a scorecard yields a score, not multiple measures of different dimensions like yards rushing and passing. These latter drivers of performance affect who wins and who loses, but they do not themselves distinguish the winner.

To reiterate, the Balanced Scorecard does not yield a score that would allow us to distinguish winners from losers. For this reason, the system is best described not as a scorecard, but as a dashboard or instrument panel. It can tell managers many interesting things about their business, but it does not give a score for the organization's performance, or even for the performance of its business units. As a senior manager of a large financial institution that spent considerable time implementing a balanced scorecard system explained to me: "We never figured out how to use the scorecard to measure performance. We used it to transfer information, a lot of information, from the divisions to the senior management team. At the end of the day, however, your performance depended on your ability to meet your targets for contribution to bottom-line profits."

Thus, because of the lack of a way for managers to think through the difficult task of determining an unambiguous performance measure in the Balanced Scorecard system, the

19. See Robert S. Kaplan and David P. Norton, "The Balanced Scorecard—Measures That Drive Performance," *Harvard Business Review*, no. January-February 1992: pp. 71-79; and Robert Kaplan and David P. Norton, *The Balanced Scorecard*. Boston, MA: Harvard Business School Press, 1996.

20. Kaplan and Norton (1996), p.8.

21. Ibid., p.10 and p.150, emphasis in original

22. Ibid., p.162

23. Ibid., p.222

result in this case was a fallback to a single and inadequate financial measure of performance (in this case, accounting profits)—the very approach that Kaplan and Norton properly wish to change. The lack of a single one-dimensional measure by which an organization or department or person will score their performance means these units or people cannot make purposeful decisions. They cannot do so because if they do not understand the tradeoffs between the multiple measures, they cannot know whether they are becoming better off (except in those rare cases when all measures are increasing in some decision).

In sum, the appropriate measure for the organization is value creation, the change in the market value of all claims on the firm. And for those organizations that wish a "flow" measure of value creation on a quarterly or yearly basis, I recommend Economic Value Added (EVA). But I hasten to add that, as the performance measures are cascaded down through the organization, neither value creation nor the year-to-year measure, EVA, is likely to be the proper performance measure at all levels. To illustrate this point, let's now look briefly at performance measurement for business units.

Measuring Divisional Performance

The proper measure for any person or business unit in a multi-divisional company will be determined mainly by two factors: the company's strategy and the actions that the person or division being evaluated can take to contribute to the success of the strategy. There are two general ways in principle that this score or objective can be determined: a centralized way and a decentralized way.

To see this let us begin by distinguishing clearly between the measure of performance (single dimensional) for a unit or person, and the drivers that the unit or person can use to affect the performance measure. In the decentralized solution, the organization determines the appropriate performance measure for the unit, and it is the person or unit's responsibility to figure out what the performance drivers are, how they influence performance, and how to manage them. The distinction here is the difference between an outcome (the performance measure) and the inputs or decision variables (the management of the performance drivers). And managers at higher levels in the hierarchy may be able to help the person or unit to understand what the drivers are and how to manage them. But this help can only go so far because the specific knowledge regarding the drivers will generally lie not in headquarters, but in the operating units. Therefore, in the end it is the accountable party, not headquarters, who will generally have the relevant specific knowledge and therefore must determine the drivers, their changing relation to results, and how to manage them.

At the opposite extreme is the completely centralized solution in which headquarters will determine the performance measure by giving the functional form to the unit that lists the drivers and describes the weight that each driver receives in the determination of the performance measure. The performance for a period is then determined by calculating the weighted average of the measures of the drivers for the period.[24] This solution effectively transfers the job of learning how to create value at all levels in the organization to the top managers, and leaves the operating managers only the job of managing the performance drivers that have been dictated to them by top management. The problem with this approach, however, is that it is likely to work only in a fairly narrow range of circumstances—those cases where the specific knowledge necessary to understand the details of the relation between changes in each driver and changes in the performance measure lies higher in the hierarchy. Although this category may include a number of very small firms, it will rule out most larger, multidivisional companies, especially in today's rapidly changing business environment.

Closing Thoughts on the Balanced Scorecard and Value Maximization

In summary, the Kaplan-Norton Balanced Scorecard is a tool to help managers understand what creates value in their business. As such, it is a useful analytical tool, and I join with Kaplan and Norton in urging managers to do the hard work necessary to understand what creates value in their organization and how to manage those value drivers. As they put it,

> *…[A] properly constructed Balanced Scorecard should tell the story of the business unit's strategy. It should identify and make explicit the sequence of hypotheses about the cause-and-effect relationships between outcome measures and the performance drivers of those outcomes. Every measure selected for a Balanced Scorecard should be an element in a chain of cause-and-effect relationships that communicates the meaning of the business unit's strategy to the organization.*[25]

But managers are almost inevitably led to try to use the multiple measures of the Balanced Scorecard as a performance measurement system. And as a performance measurement system, the Balanced Scorecard will lead to confusion, conflict, inefficiency, and lack of focus. This is bound to happen as operating managers guess at what the tradeoffs might be between each of the dimensions of performance. And this uncertainty will generally lead to conflicts with managers at headquarters, who are likely to have different assessments of the tradeoffs. Such conflicts, besides causing disappointments and confusion about operating decisions, could also lead to attempts by operat-

24. And of course I do not mean to imply that the functional relationship between the value drivers and the performance measure will always be a simple weighted average. Indeed, in general it will be more complicated than this.

25. Ibid., p.31

ing managers to game the system—by, say, performing well on financial measures while sacrificing nonfinancial ones. Moreover, there is no logical or principled resolution of the resulting conflicts unless all the parties come to agreement about what they are trying to accomplish; and this means specifying how the score is calculated—in effect, figuring out how the balance in the Balanced Scorecard is actually attained.

As we saw earlier, even if it were possible to come up with a truly "optimizing" system where all the weights and the tradeoffs among the multiple measures and drivers were specified—a highly doubtful proposition—reaching agreement between headquarters and line management over the proper weighting of the measures and their linkage to the corporate reward system would be an enormously difficult, if not an impossible, undertaking. In addition, it would surely be impossible to keep the system continuously updated so as to reflect all the changes in a dynamic local and worldwide competitive landscape.

A 1996 survey of Balanced Scorecard implementations by Towers Perrin gives a fairly clear indication of the problems that are likely to arise with it.[26] Perhaps most troubling, 70% of the companies using a scorecard also reported using it for compensation—and an additional 17% were considering doing so. And, not surprisingly, 40% of the respondents said they believed that the large number of measures weakened the effectiveness of the measurement system. What's more, in their empirical test of the effects of the balanced scorecard implementation in a global financial services firm, a 1997 study by Christopher Ittner, David Larcker, and Marshall Meyer concluded that the first issue their study raises for future research is "defining precisely what 'balance' is and the mechanisms through which 'balance' promotes performance."[27] As I have argued in this paper, this question cannot be answered because "balance" is a term used by Balanced Scorecard advocates as a substitute for thorough analysis of one of the more difficult parts of the performance measurement system—the necessity to evaluate and make tradeoffs. They and others have been seduced by this hurrah word (who can argue for "unbalanced"?) into avoiding careful thought on the issues.

In fact, the sooner we get rid of the word "balance" in these discussions, the better we will be able to sort out the solutions. Balance cannot ever substitute for having to deal with the difficult issues associated with specifying the tradeoffs among multiple goods and bads that determine the overall score for an organization's success. We must do this to stand a chance of creating an organizational scoreboard that actually gives a score—which is something every good scoreboard must do.

Closing Thoughts on Stakeholder Theory

Stakeholder theory plays into the hands of special interests that wish to use the resources of corporations for their own ends. With the widespread failure of centrally planned socialist and communist economies, those who wish to use nonmarket forces to reallocate wealth now see great opportunity in the playing field that stakeholder theory opens to them. Stakeholder theory gives them the appearance of legitimate political access to the sources of decision-making power in organizations, and it deprives those organizations of a principled basis for rejecting those claims. The result is to undermine the foundations of value-seeking behavior that have enabled markets and capitalism to generate wealth and high standards of living worldwide.

If widely adopted, stakeholder theory will reduce social welfare even as its advocates claim to increase it—much as happened in the failed communist and socialist experiments of the last century. And, as I pointed out earlier, stakeholder theorists will often have the active support of managers who wish to throw off the constraints on their power provided by the value-seeking criterion and its enforcement by capital markets, the market for corporate control, and product markets. For example, stakeholder arguments played an important role in persuading the U.S. courts and legislatures to limit hostile takeovers through legalization of poison pills and state control shareholder acts. And we will continue to see more political action limiting the power of these markets to constrain managers. In sum, special interest groups will continue to use the arguments of stakeholder theory to legitimize their positions, and it is in our collective interest to expose the logical fallacy of these arguments.

26. Towers Perrin, "Inside 'the balanced scorecard'," *Compuscan Report*, January 1996: pp. 1-5.

27. Ittner, Cristopher, David F. Larcker, and Marshal W. Meyer, "Performance, Compensation, and the Balanced Scorecard," Unpublished, Wharton School, U. of Pennsylvania, November 1, 1997.

Journal of Applied Corporate Finance (ISSN 1078-1196 [print], ISSN 1745-6622 [online]) is published quarterly, on behalf of Morgan Stanley by Wiley Subscription Services, Inc., a Wiley Company, 111 River St., Hoboken, NJ 07030-5774. Postmaster: Send all address changes to JOURNAL OF APPLIED CORPORATE FINANCE Journal Customer Services, John Wiley & Sons Inc., 350 Main St., Malden, MA 02148-5020.

Information for Subscribers *Journal of Applied Corporate Finance* is published in four issues per year. Institutional subscription prices for 2010 are:

Print & Online: US$416 (US), US$499 (Rest of World), €323 (Europe), £255 (UK). Commercial subscription prices for 2010 are: Print & Online: US$556 (US), US$663 (Rest of World), €429 (Europe), £338 (UK). Individual subscription prices for 2010 are: Print & Online: US$105 (US), £59 (Rest of World), €88 (Europe), £59 (UK). Student subscription prices for 2010 are: Print & Online: US$37 (US), £21 (Rest of World), €32 (Europe), £21 (UK).

Prices are exclusive of tax. Asia-Pacific GST, Canadian GST and European VAT will be applied at the appropriate rates. For more information on current tax rates, please go to www3.interscience.wiley.com/about us/journal_ordering_and_payment.html#Tax. The price includes online access to the current and all online back files to January 1997, where available. For other pricing options, including access information and terms and conditions, please visit www.interscience.wiley.com/journal-info.

Journal Customer Services:For ordering information, claims and any enquiry concerning your journal subscription please go to interscience.wiley.com/support or contact your nearest office.
Americas: Email: cs-journals@wiley.com; Tel: +1 781 388 8598 or +1 800 835 6770 (toll free in the USA & Canada).
Europe, Middle East and Africa: Email: cs-journals@wiley.com; Tel: +44 (0) 1865 778315.
Asia Pacific: Email: cs-journals@wiley.com; Tel: +65 6511 8000.
Japan: For Japanese speaking support, Email: cs-japan@wiley.com; Tel: +65 6511 8010 or Tel (toll-free): 005 316 50 480. Further Japanese customer support is also available at www.interscience.wiley.com/support
Visit our Online Customer Self-Help available in 6 languages at www.interscience.wiley.com/support

Production Editor: Joshua Gannon (email:jacf@wiley.com).
Delivery Terms and Legal Title Prices include delivery of print journals to the recipient's address. Delivery terms are Delivered Duty Unpaid (DDU); the recipient is responsible for paying any import duty or taxes. Legal title passes to the customer on despatch by our distributors.

Back Issues Single issues from current and recent volumes are available at the current single issue price from cs-journals@wiley.com. Earlier issues may be obtained from Periodicals Service Company, 11 Main Street, Germantown, NY 12526, USA. Tel: +1 518 537 4700, Fax: +1 518 537 5899, Email: psc@periodicals.com

This journal is available online at Wiley InterScience. Visit www.interscience.wiley.com to search the articles and register for table of contents e-mail alerts.

Access to this journal is available free online within institutions in the developing world through the AGORA initiative with the FAO, the HINARI initiative with the WHO and the OARE initiative with UNEP. For information, visit www.aginternetwork.org, www.healthinternetwork.org, www.healthinternetwork.org, www.oarescience.org, www.oarescience.org

Wiley's Corporate Citizenship initiative seeks to address the environmental, social, economic, and ethical challenges faced in our business and which are important to our diverse stakeholder groups. We have made a long-term commitment to standardize and improve our efforts around the world to reduce our carbon footprint. Follow our progress at www.wiley.com/go/citizenship

Abstracting and Indexing Services
The Journal is indexed by Accounting and Tax Index, Emerald Management Reviews (Online Edition), Environmental Science and Pollution Management, Risk Abstracts (Online Edition), and Banking Information Index.